

SoS' 26 - P06 Computational Physics

GPU-Accelerated VQE Quantum Chemistry Simulator - PoA

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Weeks

Week 1: Research the physics behind the Variational Quantum Eigensolver (VQE) algorithm and learn how basic quantum logic gates operate using linear algebra.

Week 2: Learn how to represent quantum states mathematically and write the core, single-qubit simulator in C++.

Week 3: Add parameterized rotation gates to the C++ code and write the functions needed to calculate the total energy of a given quantum state.

Week 4: Learn the basics of Qiskit to build identical circuits in Python. Use this to debug the single-threaded CPU version by comparing the expected energy calculations against Qiskit's simulator.

→ **MIDTERM REPORT**

Week 5: Start learning CUDA and set up the NVIDIA GPU environment to prepare for parallelizing the heavy matrix calculations.

Week 6: Implement the parallelization code to run the quantum simulator on the dedicated NVIDIA GPU and test it with multiple qubits at maximal capacity.

Week 7: Write a Python script using an open-source chemistry tool (like OpenFermion or Qiskit Nature) to generate the Hamiltonian for a Hydrogen molecule. Connect this script to the C++ simulator to automatically find the ground state energy.

Week 8: Conduct final debugging, benchmark the CPU vs. GPU speeds, and compare the custom engine's final VQE results directly against Qiskit's built-in VQE solver.

→ **ENDTERM REPORT**

References

1. *Quantum Computation and Quantum Information* by M.A. Nielsen and I.L. Chuang (Chapters 2 & 4 for Linear Algebra and Quantum Circuits).
2. IBM Qiskit Online Textbook (Section on the Variational Quantum Eigensolver).
3. Peruzzo, A., McClean, J., Shadbolt, P., et al. (2014). *A variational eigenvalue solver on a photonic quantum processor*. Nature Communications, 5, 4213.